PHYSIOLOGY OF FLUIDS

Body compartments and Water distribution



Universidad de Navarra

Nicolas Varela

31st October 2025



- « Ce qu’est réellement la médecine: la haine de la maladie et l’amour du malade ».
- « What Medicine really is: hate the sickness and love the sick ».

Jérôme Lejeune. 1961

Universidad de Navarra

Summary

- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartments composition
- Diffusion vs. Transport
- Compartmental separation

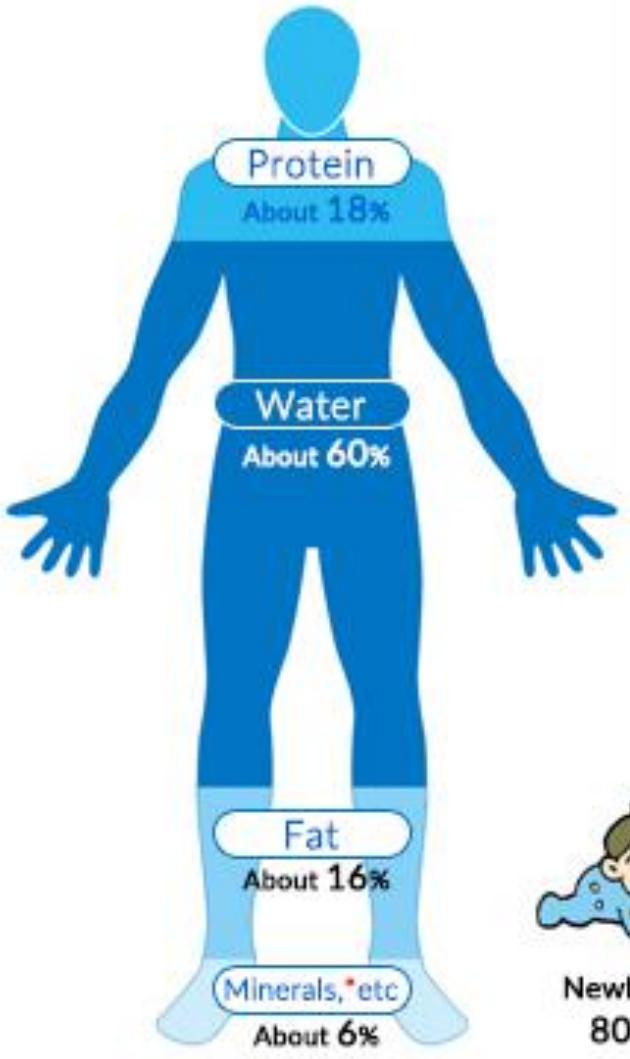
Universidad de Navarra

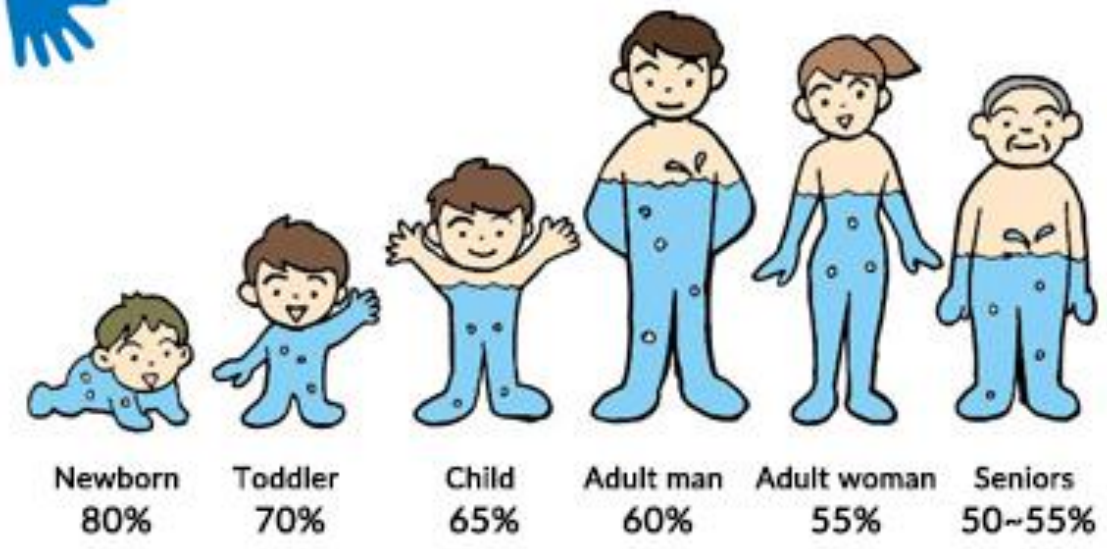
Summary

- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartments composition
- Diffusion vs. Transport
- Compartmental separation

Universidad de Navarra

Body water proportion





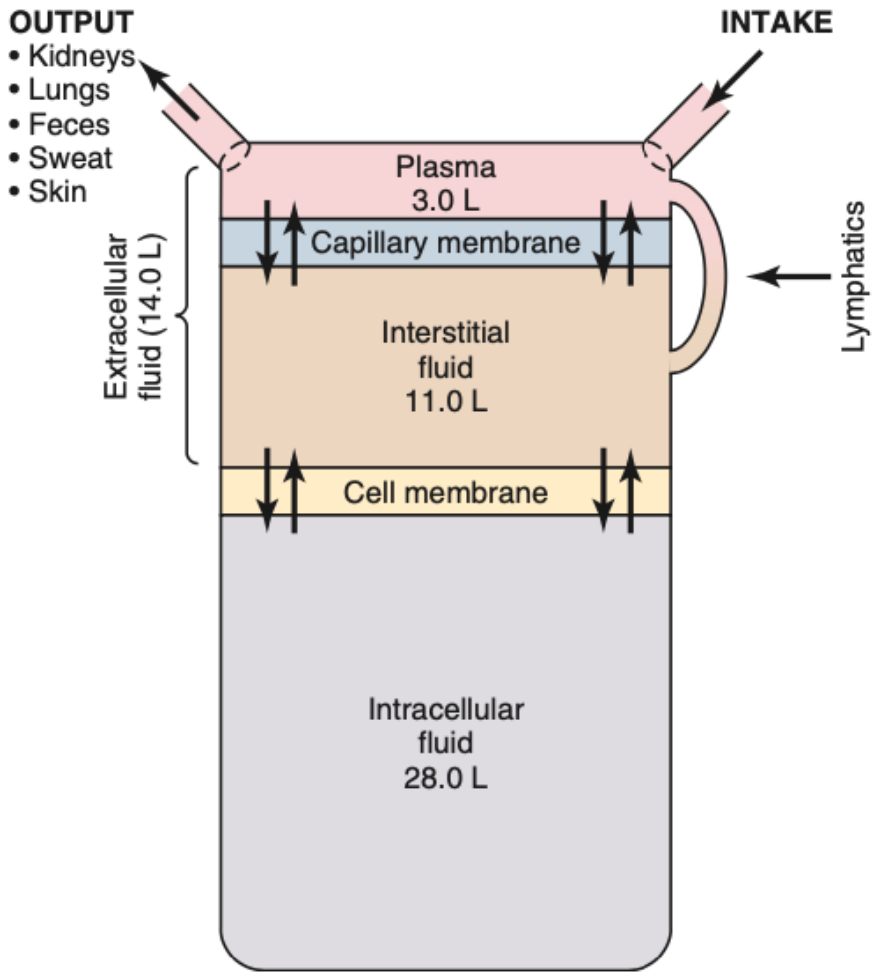
Universidad de Navarra

Daily intake and output of water

	Normal	Prolonged, Heavy Exercise
Intake		
Fluids ingested	2100	?
From metabolism	200	200
Total intake	2300	?
Output		
Insensible: skin	350	350
Insensible: lungs	350	650
Sweat	100	5000
Feces	100	100
Urine	1400	500
Total output	2300	6600

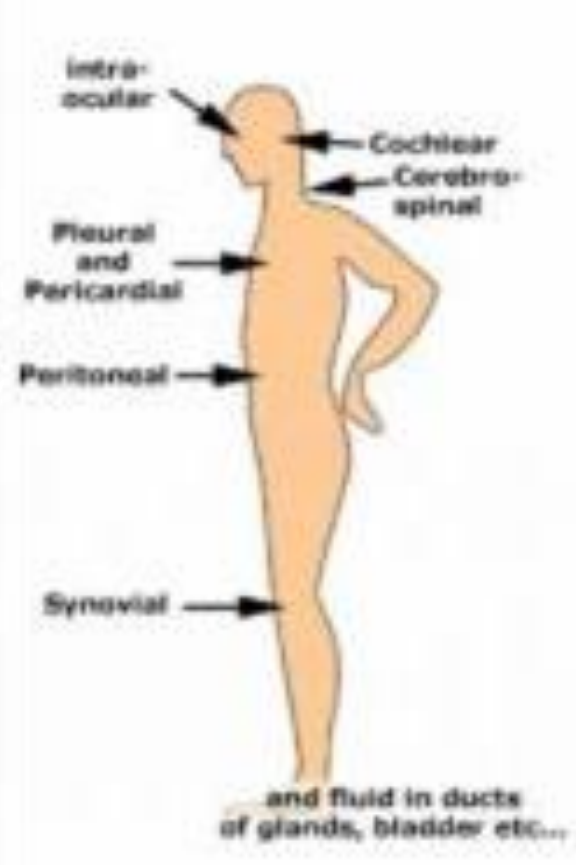
Body fluid compartments

- Two main compartments
  - ICF: 28L
  - ECF: 14L
    - Interstitial: 11L
    - Plasma: 3L
- Blood volume
- Third compartment: transcellular



Transcellular fluid

- Also called third space
- Neither ICF, nor ECF
- Not connected to blood stream
- Important in pathological conditions



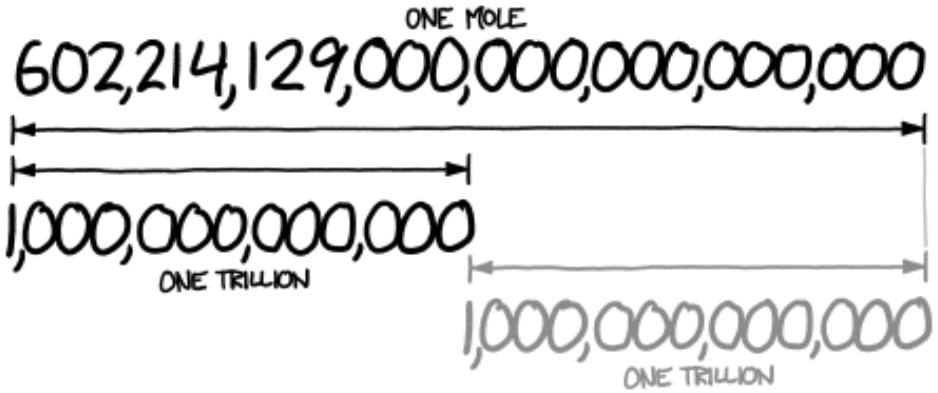
Universidad de Navarra

Summary

- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartments composition
- Diffusion vs. Transport
- Compartmental separation

Physical Principles

- Mole: 6,022 x10^{23} = 602 billion trillion

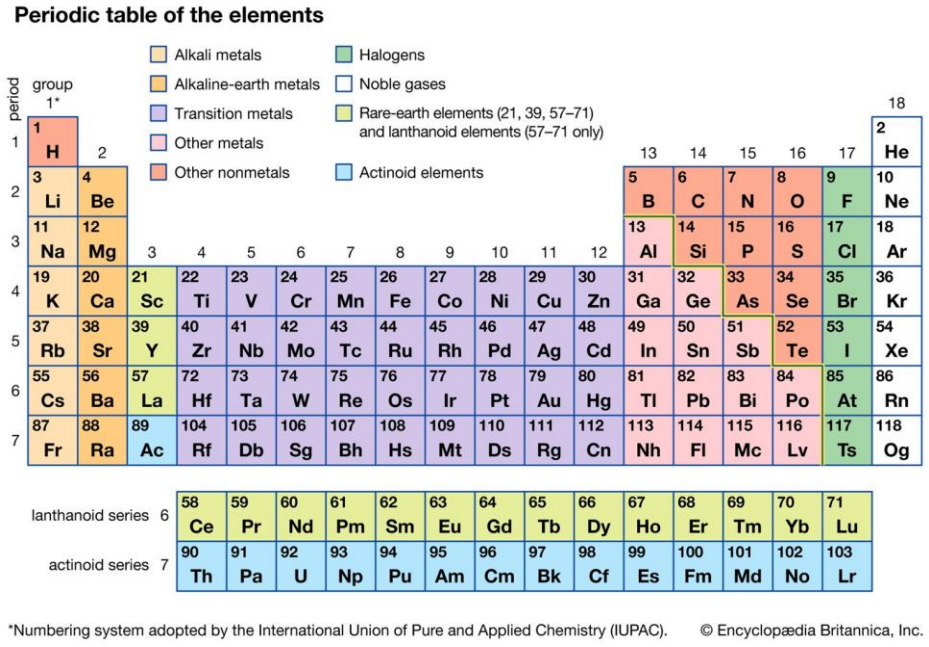


- Molecular weight = sum of atomic weights
- eg.: H_2O: 2x1 + 1x16=18g/mol

- Molarity: moles/Litre of solution
- Molality: moles/Kg of solvent

- Osmolarity: Osmoles/L

- Milliequivalents: millimoles x ionic charge



Universidad de Navarra

What is the difference between Molarity and Osmolarity?

- Molarity means the number of moles of solute particles per unit volume of the solution, but osmolarity means the number of osmoles of solute particles per unit volume of the solution.
- Unit of molarity is $mol.dm^{-3}$ whereas unit of osmolarity is Osm/L.
- When a compound cannot dissociate when dissolved, osmolarity and molarity of that compound will be similar (eg. Glucose), but if the compound dissociates they will be different (1 mole of NaCl gives 2 Osm of diluted particles; 1 for Na^+, 1 for Cl^-)

- Molarity is the concentration *per se* while Osmolarity is related to the osmotic concentration

Universidad de Navarra

What is the difference between Osmolarity and milliEquivalents?

- Osmolarity is related to the osmotic pressure, while mEq take into account the charge.

- Osmolarity is about number of ions, mEq is about charge.

- 1mOsmol Na^+ equals 1mEq of Na^+
- 1mOsmol Ca^{2+} equals 2mEq of Ca^{2+}

- 1mM NaCl = 2mOsm = 2mEq
- 1mM CaPr = 2mOsm = 4mEq

Universidad de Navarra

Summary

- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartmental composition
- Diffusion vs. Transport
- Compartmental separation

Universidad de Navarra

Fluid composition

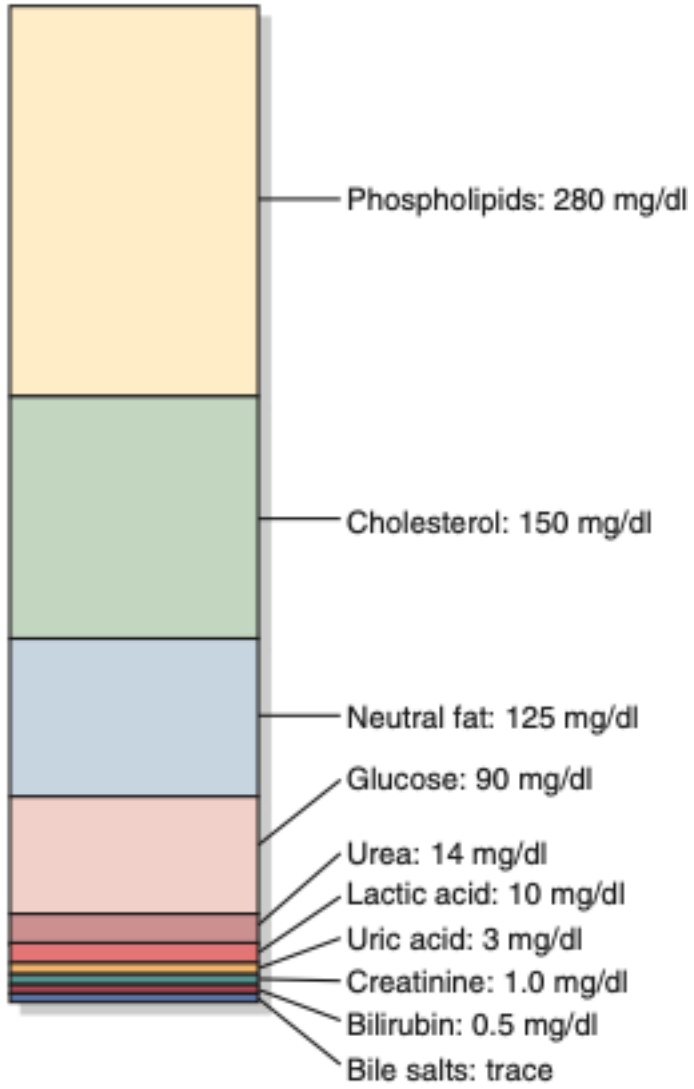
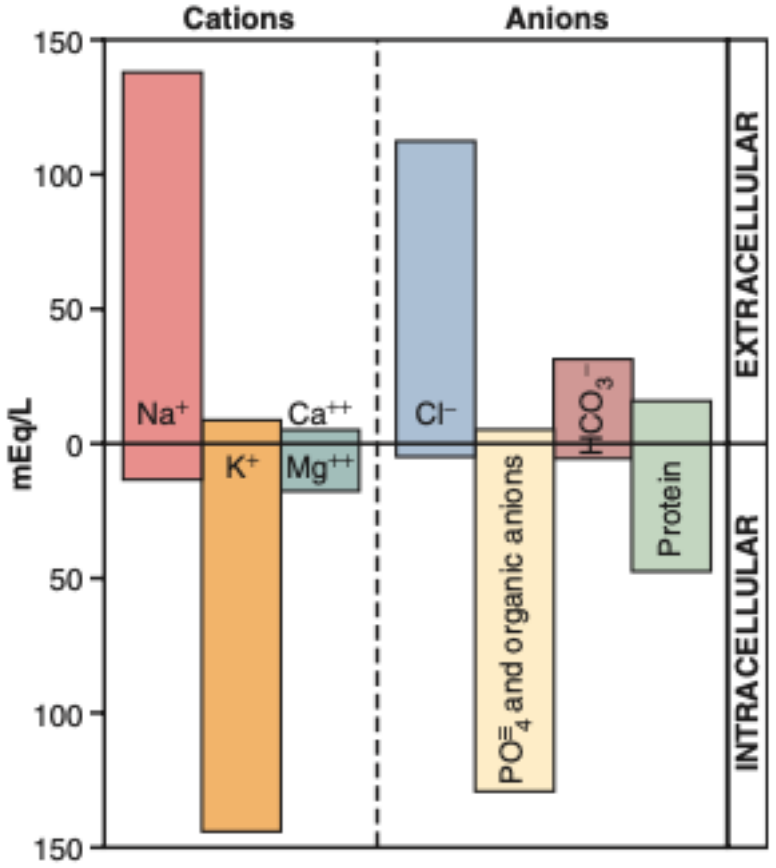


Figure 25-3. Nonelectrolytes of the plasma.



Universidad de Navarra

Ionic composition

- Interstitial and plasma are similar
  - Donnan effect

- ECF: Na^+, Ca^{2+} Cl^-, HCO_3^-
- ICF: K^+, Mg^{2+}, HPO_4^-, $Prot^-$

- ICF: 4x more proteins

	Plasma (mOsm/L H_2O)	Interstitial (mOsm/L H_2O)	Intracellular (mOsm/L H_2O)
Na^+	142	139	14
K^+	4.2	4.0	140
Ca^{++}	1.3	1.2	0
Mg^{++}	0.8	0.7	20
Cl^-	106	108	4
HCO_3^-	24	28.3	10
HPO_4^-, $H_2PO_4^-$	2	2	11
SO_4^-	0.5	0.5	1
Phosphocreatine			45
Carnosine			14
Amino acids	2	2	8
Creatine	0.2	0.2	9
Lactate	1.2	1.2	1.5
Adenosine triphosphate			5
Hexose monophosphate			3.7
Glucose	5.6	5.6	
Protein	1.2	0.2	4
Urea	4	4	4
Others	4.8	3.9	10
Total mOsm/L	299.8	300.8	301.2

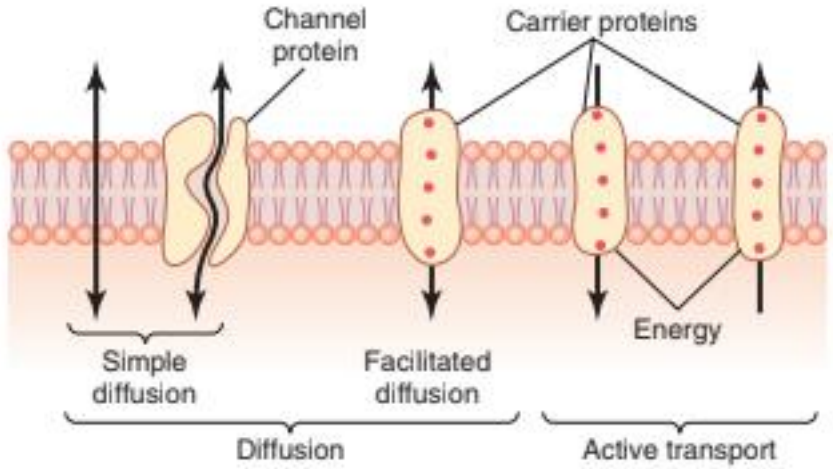
Universidad de Navarra

Summary

- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartments composition
- Diffusion vs. Transport
- Compartmental separation

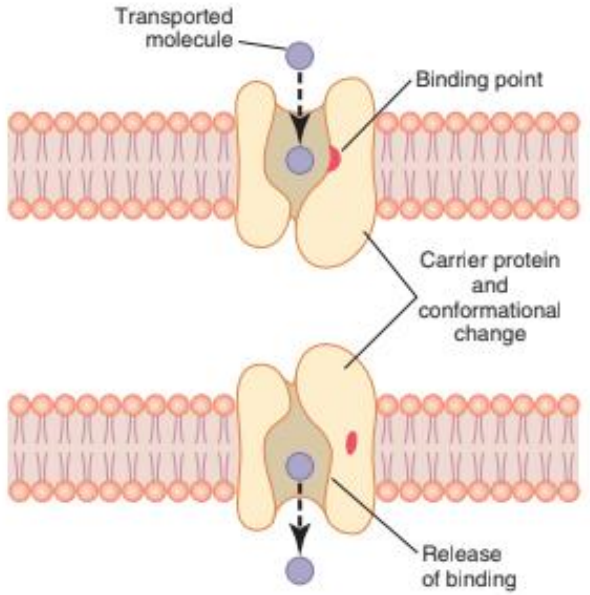
Universidad de Navarra

Diffusion/Transport



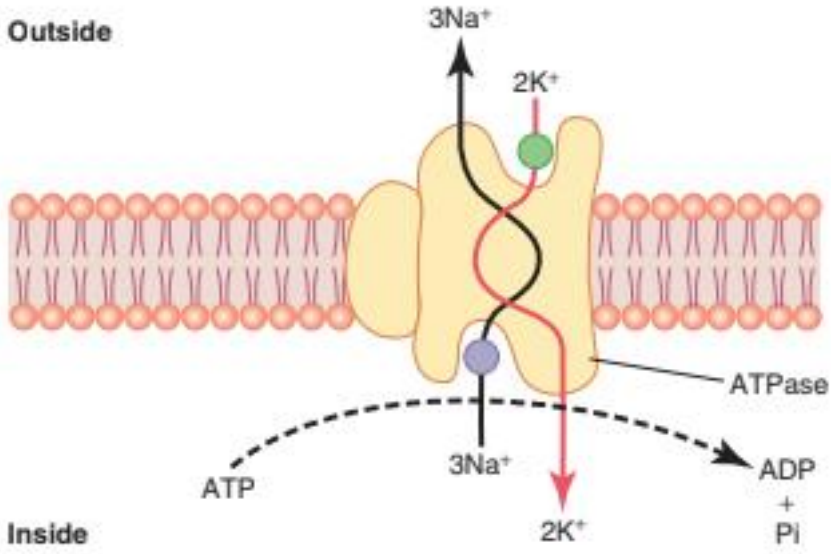
- Diffusion is passive while Transport is active
- Transport requires energy

- Diffusion: Aquaporins
- Transport: Sodium-potassium pump



Sodium-potassium pump

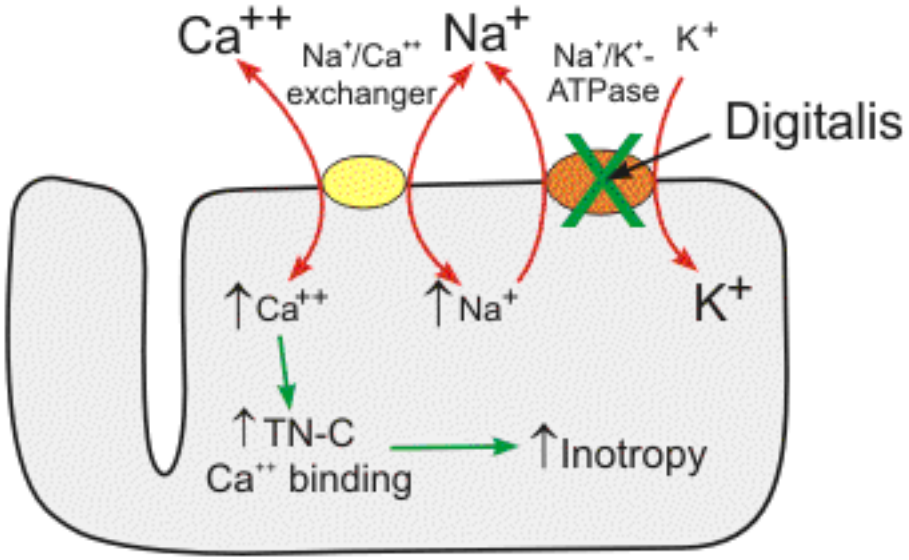
- 2 globular proteins, α and β
  - 3 binding sites for Na^+
  - 2 binding sites for K^+
  - ATPase activity



- Net effects:
  - Maintains Na^+ and K^+ concentrations
  - Maintains cellular volume
  - Creates a negative electrical voltage (basis of nerve function)

Clinical considerations

- Some molecules can activate or inhibit Na^+/K^+ pump
- Activators: Insulin
- Inhibitors: Digoxin (foxglove plant)





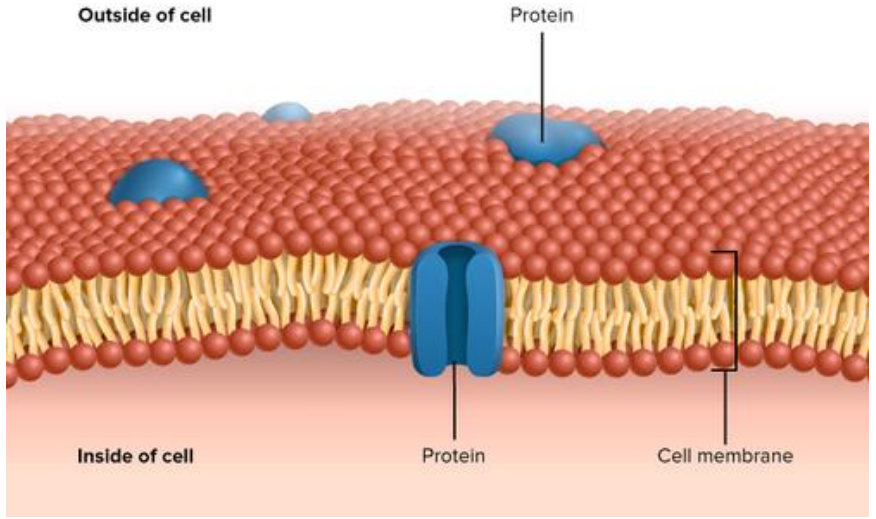
Universidad de Navarra

Summary

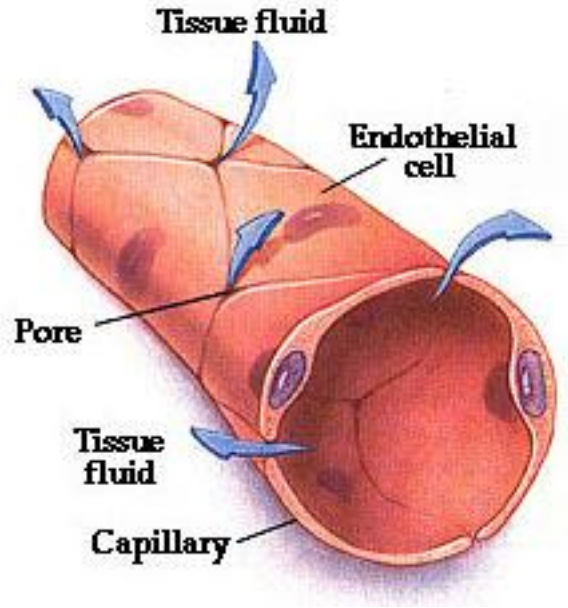
- Water compartments of the body
  - ICF
  - ECF
  - Transcellular
- Physical principles: units
- Compartments composition
- Diffusion vs. Transport
- Compartmental separation

Compartmental separation

- ICF and ECF are theoretically isolated: bilayer membrane
- Water can diffuse: aquaporins



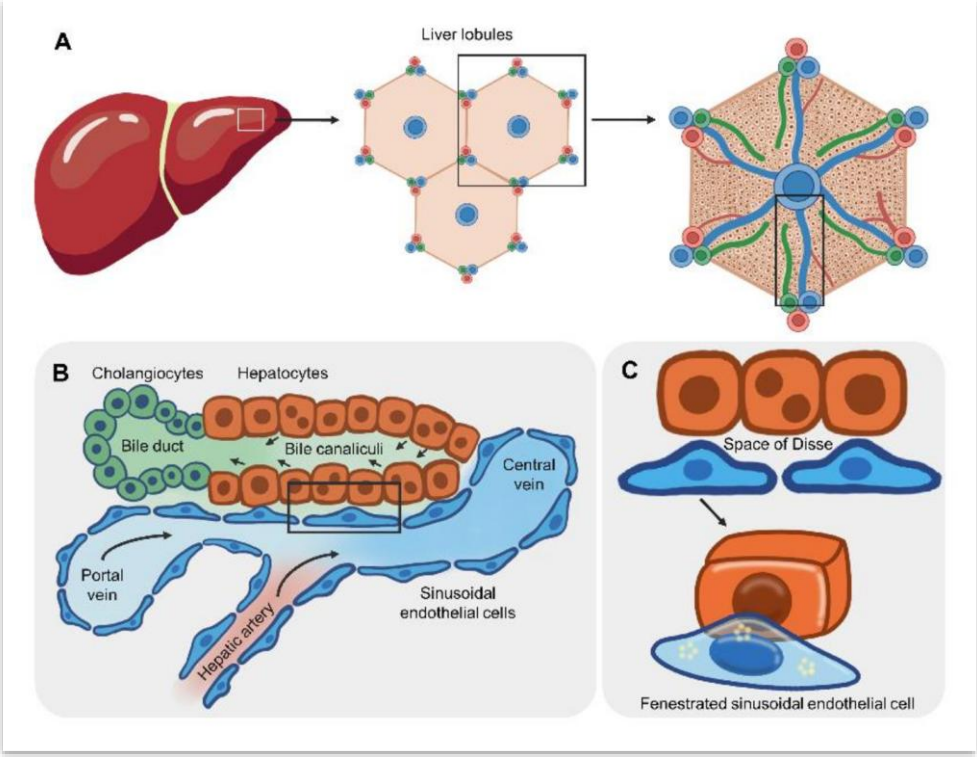
- Plasma and interstitial fluid are separated by endothelium, presenting pores (6-7nm). Only proteins can’t travel through endothelium.

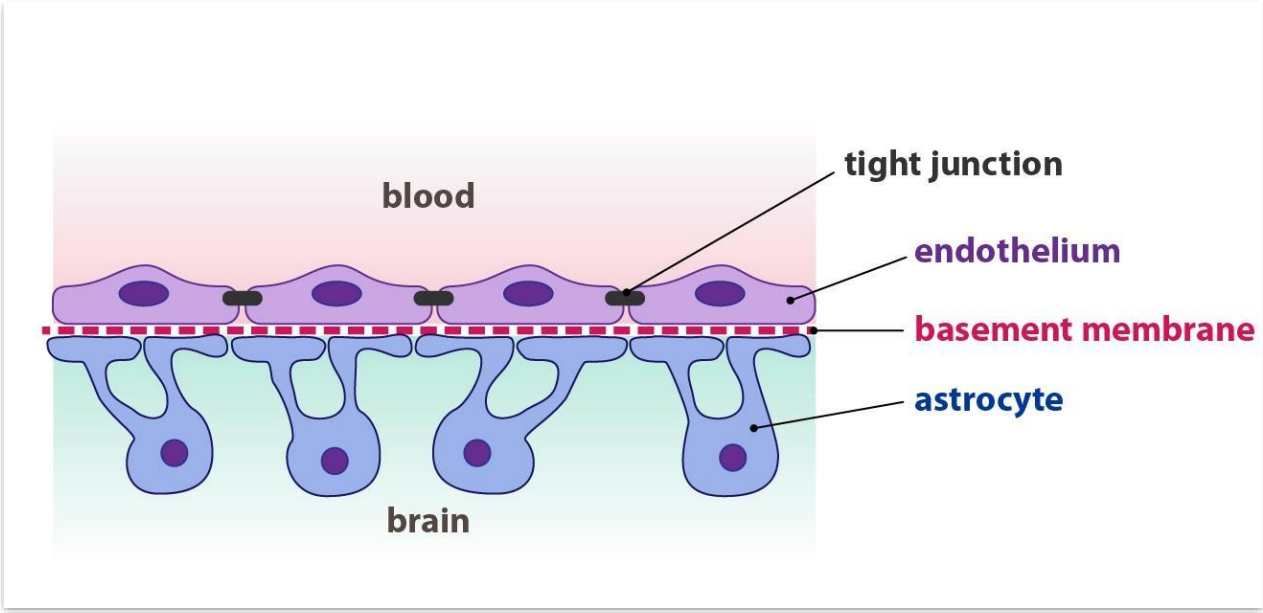


Universidad de Navarra

Exceptions

- Liver structures (fenestration)





- Blood-brain barrier (tight junction)

PHYSIOLOGY OF FLUIDS

Water diffusion



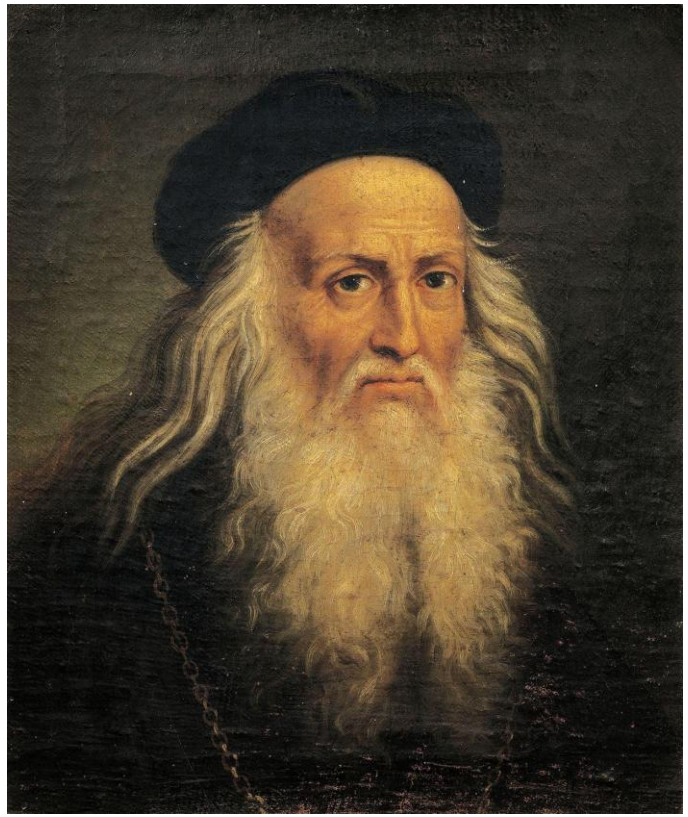
Universidad de Navarra

Nicolas Varela

31st October 2025

- “Water is the driving force of all nature.”

  Leonardo da Vinci 1452-1519



Universidad de Navarra

Summary

- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

Universidad de Navarra

Summary

- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

Universidad de Navarra

Osmolarity, osmosis and osmotic pressure

- Osmolarity is the number of osmoles of solute particles per unit volume of the solution
- Semipermeable membrane implies Osmosis; i.e. osmolarities tend to be even.
- Osmosis is the water exchange.
- Osmotic pressure (π) is the force that pushes water form one side to the other.
- Osmotic pressure depends on 4 factors: temperature, concentration of solute, Nbr of particles (n) and Staverman's coef (σ)

$$\pi = [X] \times n \times \sigma = Osm \cdot \sigma$$

Universidad de Navarra

Staverman's coef (σ)

- Staverman's reflection coefficient (σ) is a unitless constant that is specific to the permeability of a membrane to a given solute

- It can vary from 0 to 1 (1=can't travel through membrane; 0=travels freely)

Universidad de Navarra

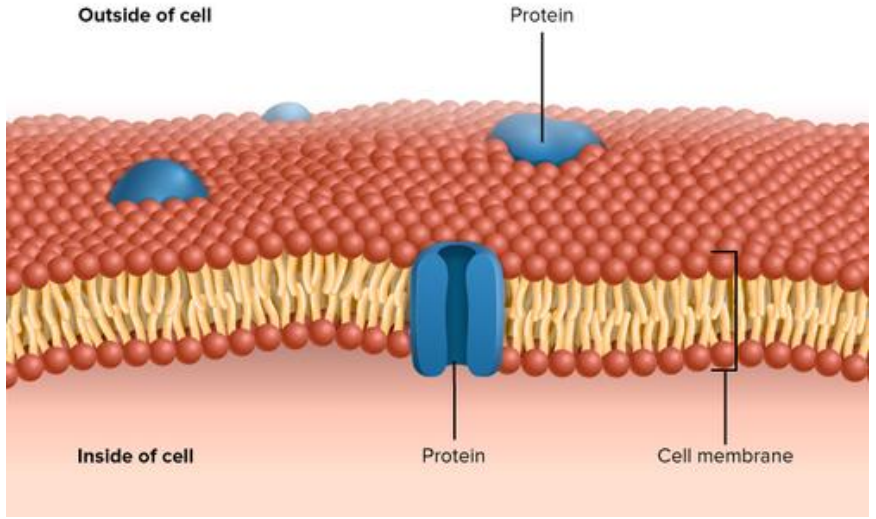
Summary

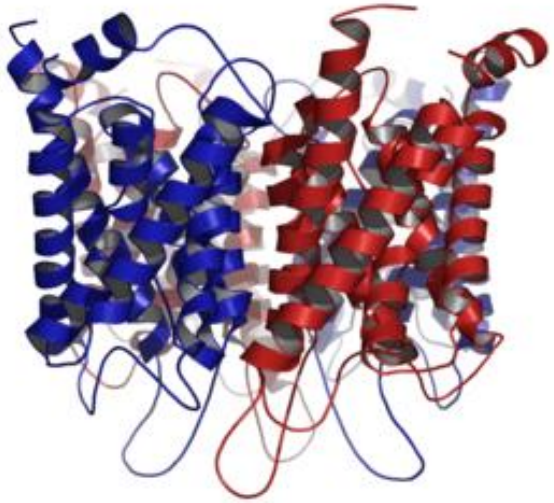
- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

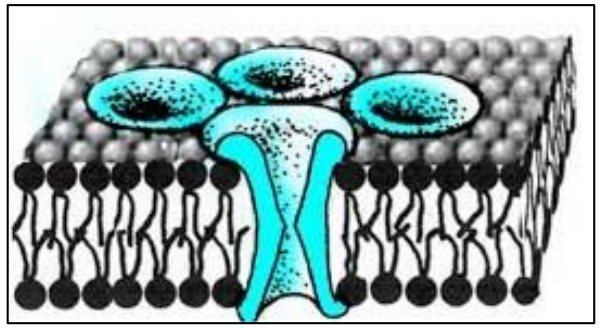
Universidad de Navarra

Aquaporins

- 13 different proteins: AQ0 -> AQ12
- 6 membrane-spanning alpha helical domains with both carboxylic and amino terminals on the cytoplasmic side
- Groups of 4
- Allows water to freely flow



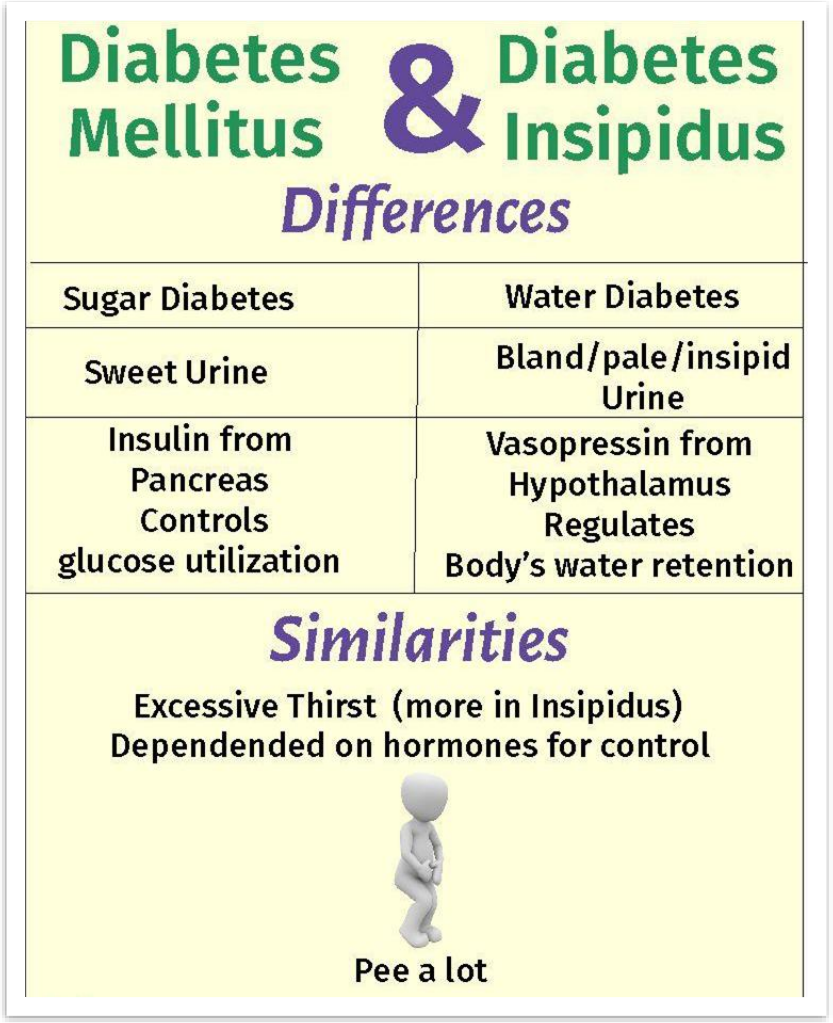




Universidad de Navarra

Clinical importance

- Mutations in the AQ2 gene cause hereditary nephrogenic diabetes insipidus in humans.
- (of note: insipidus=lack of taste, mellitus=sweetened with honey)
- DI patients:
  - up to 20L of urine production per day
  - Unable to concentrate urine (no response to vasopressin)



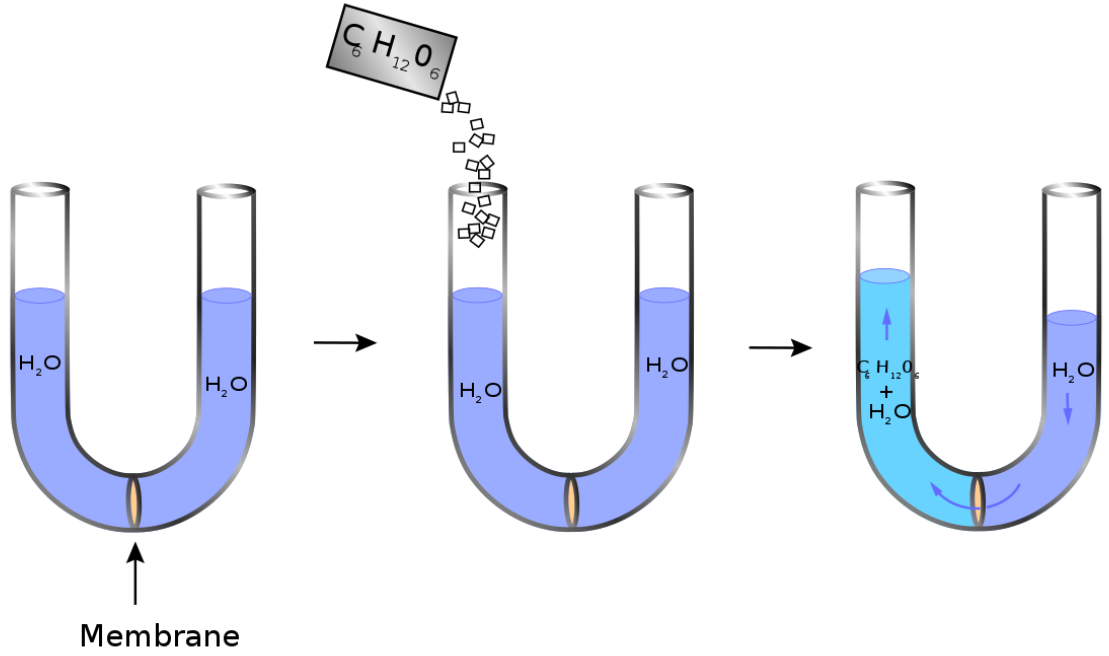
Universidad de Navarra

Summary

- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

Universidad de Navarra

Osmotic equilibrium



- Need for osmolarity equilibrium between all compartments
- ECF and ICF have the same osmolarity, due to different compositions
- Osmolarity depends on concentrations
- Water flowing freely allows this to happen

Universidad de Navarra

Plasma osmolarity

- Depends on Na^+, glucose and BUN (Blood Urea Nitrogen)
- Can be calculated by van't Hoff's law

- $$Osm_{pl} = 2 \cdot \left[Na^{+} \; in \frac{mOsm}{L}\right] + \frac{[Glu \; in \frac{mg}{dL}]}{18} + \frac{[BUN \; in \frac{mg}{dL}]}{2{,}8}$$

- $$Osm_{pl} = 2{\times}145 + \frac{100}{18} + \frac{1}{2{,}8} = 295 \text{ mOsm/L}$$

Glucose molecular weight is 180
BUN molecular weight is 28
This is divided by 10 as mg/dL compared to mOsm/L

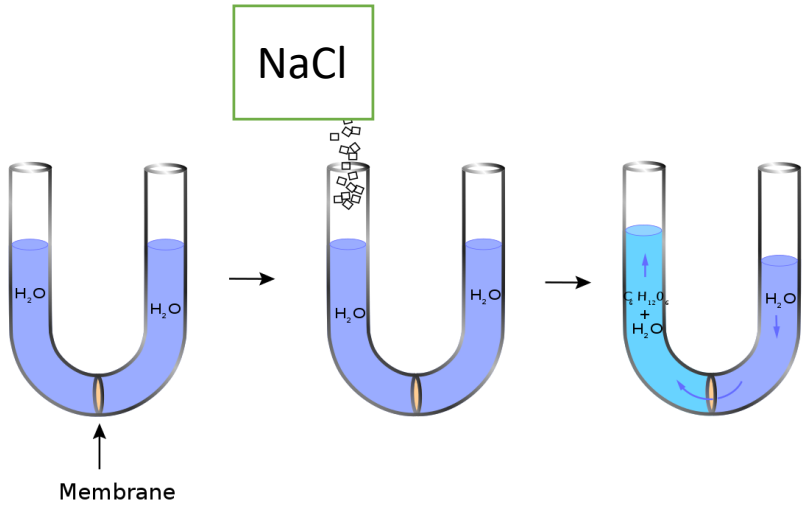
$[Na^+]$ x2 as NaCl or $NaHCO_3$

Universidad de Navarra

Impact of components in osmolarity

- As we can see, main implied ion is Na^+
- Moreover: Urea's Staverman's coeff is fairly low (hence diffuses)

- When $Na^+ \uparrow$, $Osm_{pl} \uparrow$ and water moves ICF$\Longrightarrow$ECF



Universidad de Navarra

Clinical consideration

- Normal saline (0,9% NaCl) (*Suero fisiológico*)
- Most widely used fluid therapy
- Considering NaCL molecular weight is 58g/mol:
- 0,9 %/L = 9 g/L = 0,155 mol/L = 155 mmol/L
- π = [Na] x n x σ = 155 x 1,87* x 1 = 290 mOsm/L

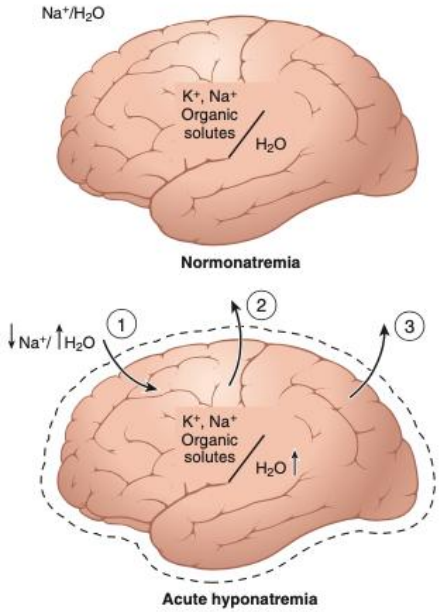
Universidad de Navarra

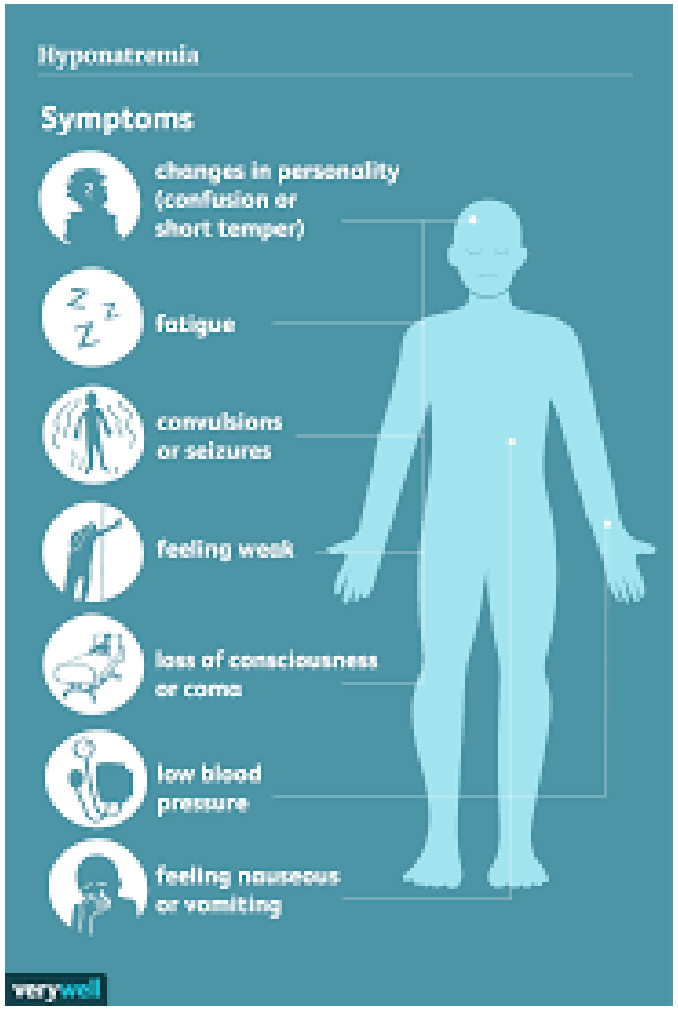
Summary

- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

What happens when Osm_{pl} ↓

- If plasma osmolarity lowers, water moves from ECF to ICF
- Problems:
  - Brain is in a closed cavity
  - When ICF↑, brain is “squished”.
  - This produces neurological symptoms:
    - Headache
    - Vomiting
    - Obnubilation
    - Seizures
    - Coma (and even death)
- Most typical cause is hyponatremia
- Hyponatremia is the most common electrolyte alteration in inpatients

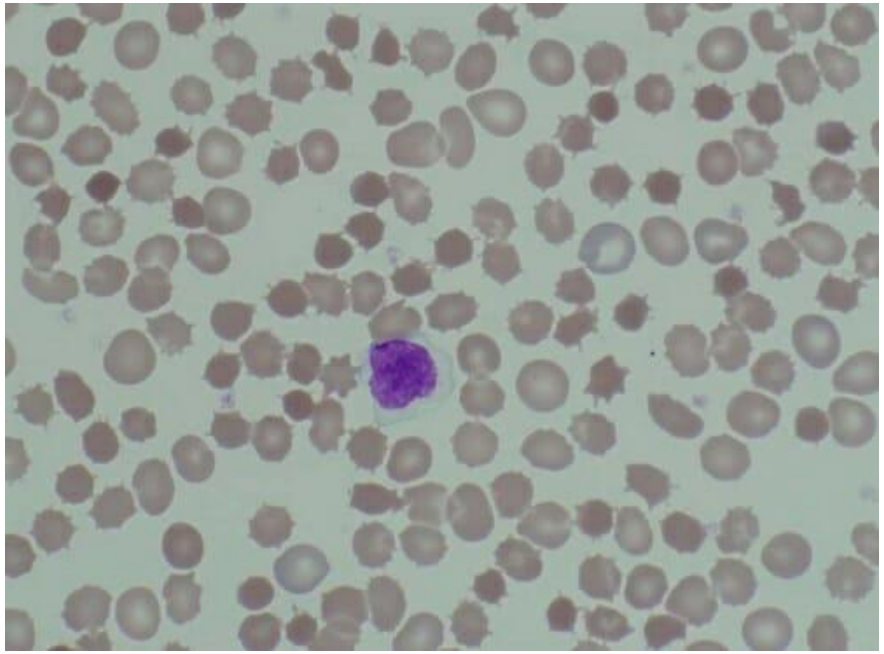




Universidad de Navarra

What happens when Osm_{pl} ↑

- If plasma osmolarity goes up, water moves from ICF to ECF
- In the brain, this could lead to increase in blood volume and potentially to a haemorrhage
- More typically, this may bay cause crenation of cells



Universidad de Navarra

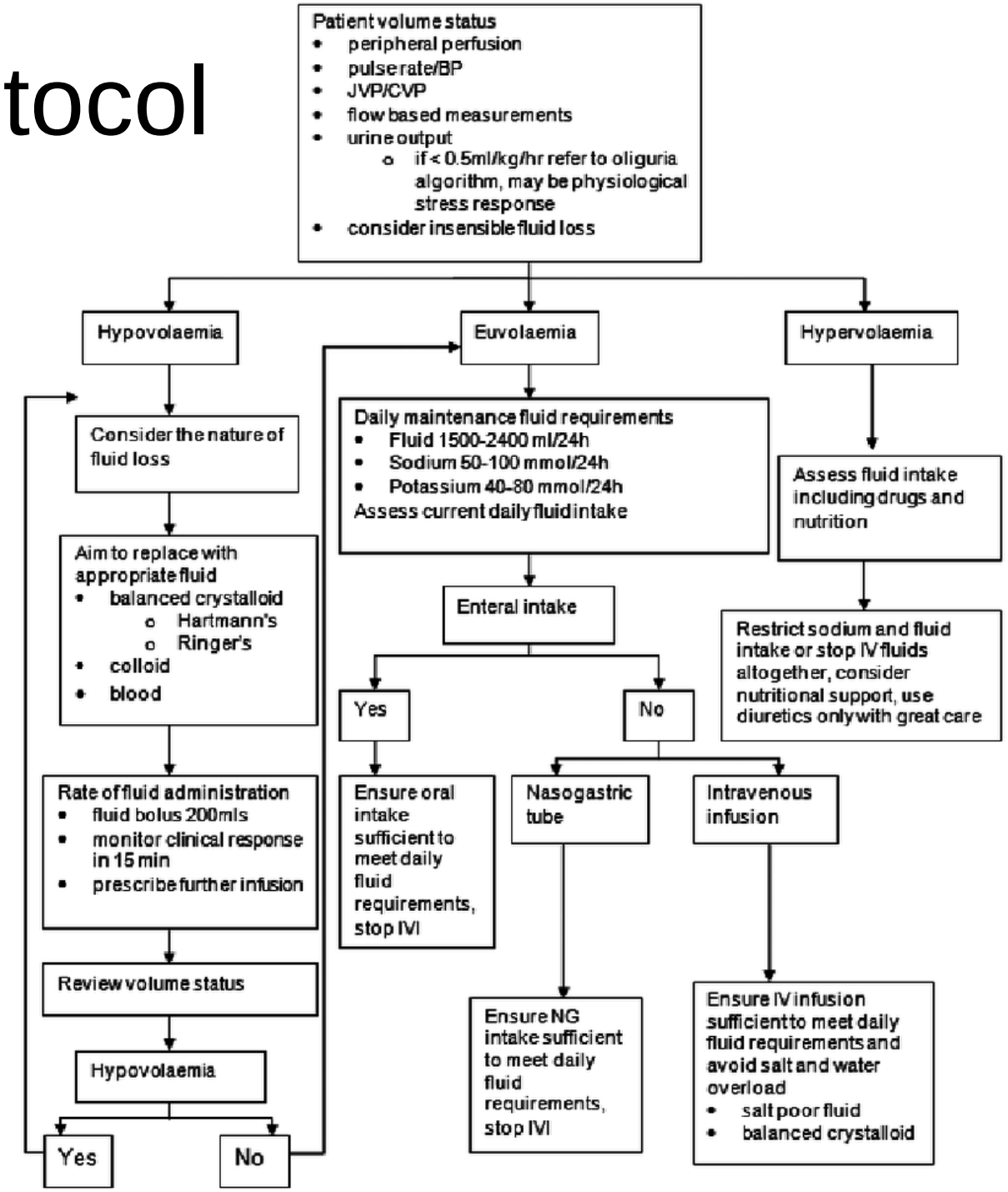
Summary

- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

Universidad de Navarra

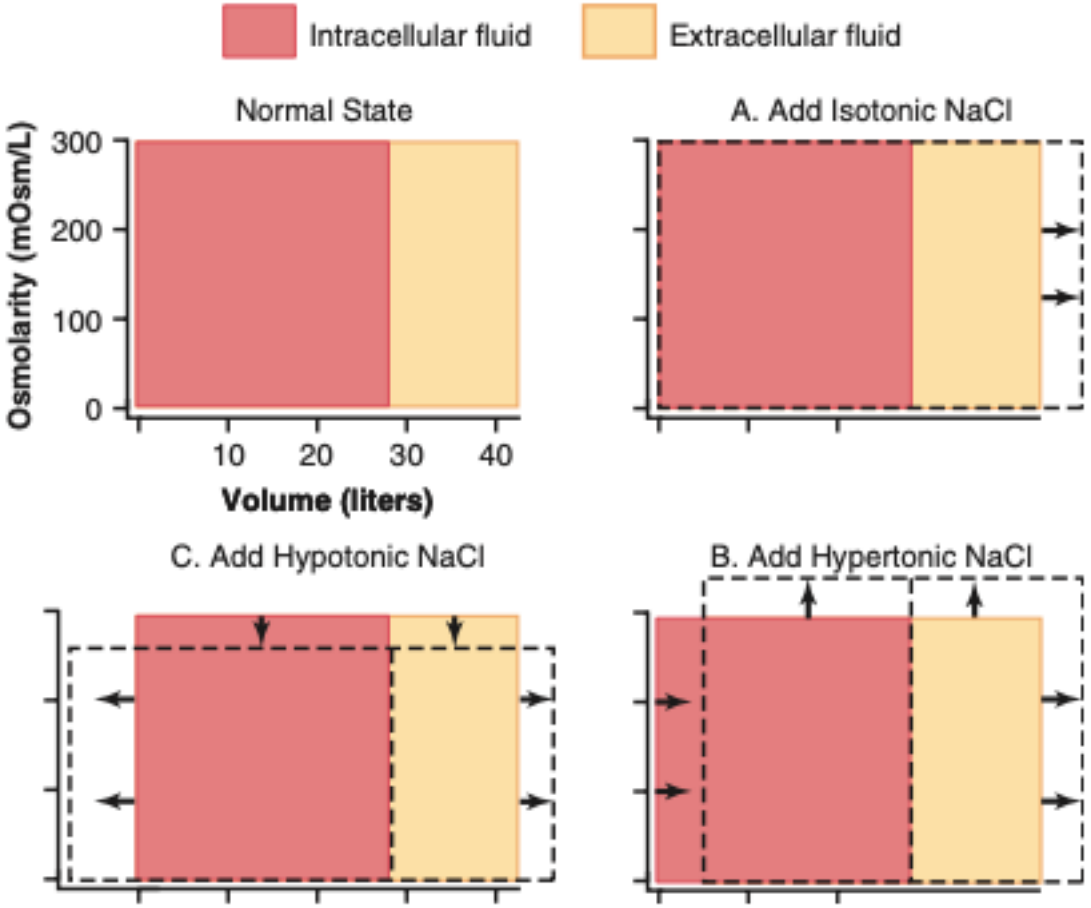
A typical protocol

- IV fluid therapy is basic
- It is the most common inpatient therapy
- Knowledge of osmosis is crucial
- It can potentially be lethal
- Euvolaemia is key



Effect of different concentrations

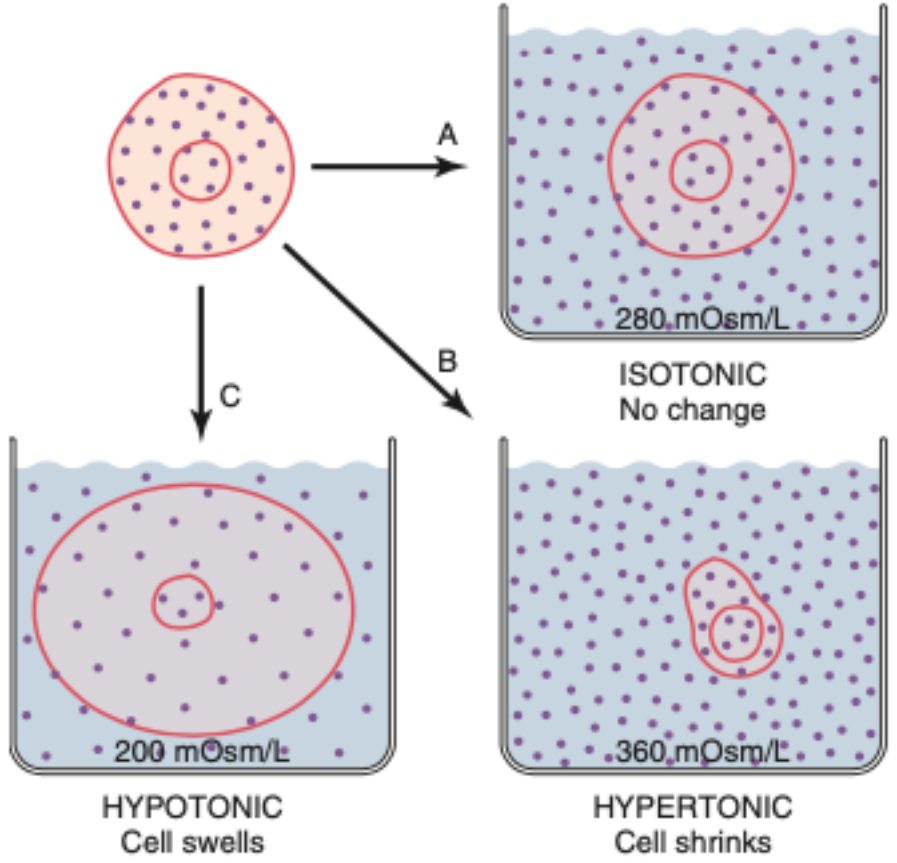
- Isotonic: 0,9% NaCl (Normal Saline)
  - 100% ECF (25% plasma, 75% interstitial)
- Hypotonic: NaCl 0,45%
  - Initially ECF
  - $Osm_{ECF}\downarrow \Longrightarrow$ Water enters ICF from ECF
  - Non-effective for Euvolaemia
- Hypertonic: 3% NaCl
  - 516mmol/L = 965 mOsm/L
  - $Osm_{ECF}\uparrow \Longrightarrow$ Water enters ECF from ICF
  - Useful in Traumatic Brain Injury



Universidad de Navarra

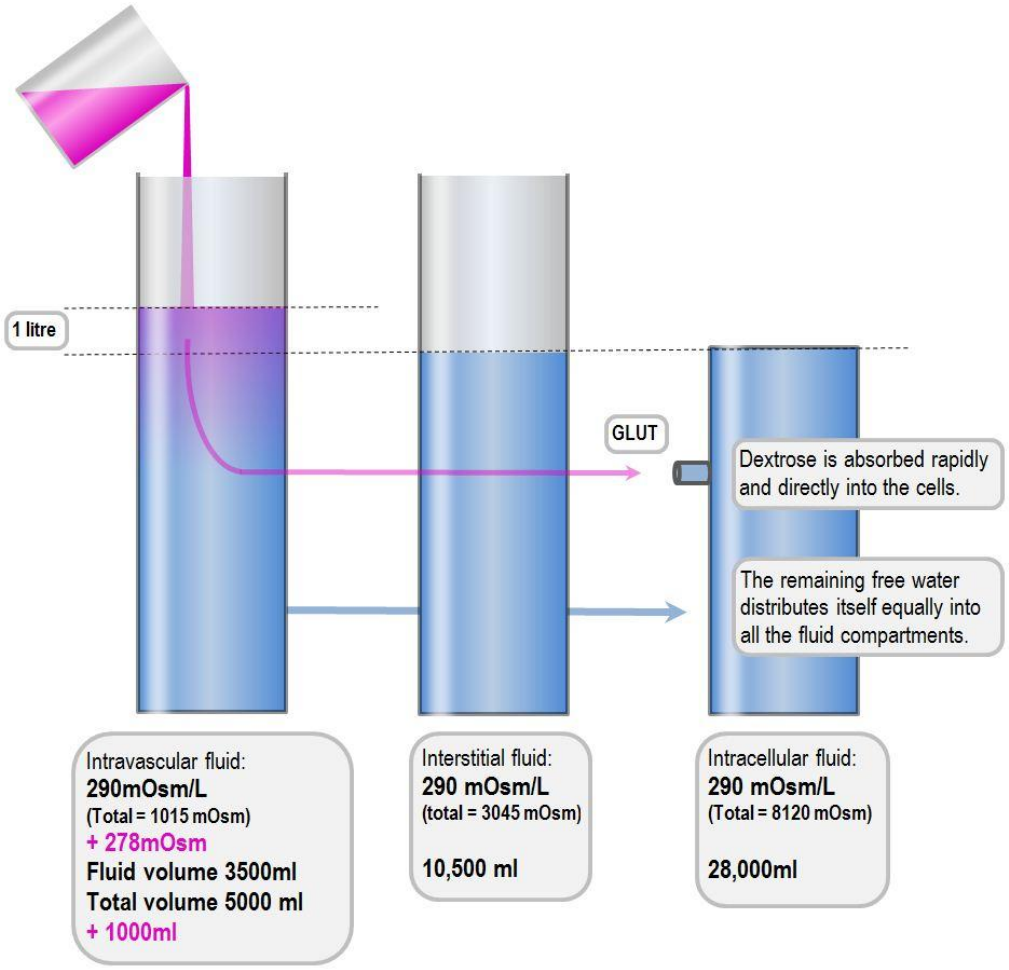
Effect on cells

- Isotonic: 0,9% NaCl (Normal Saline)
  - No change
- Hypotonic: NaCl 0,45%
  - Cells swell (and even burst)
- Hypertonic: 3% NaCl
  - Cells shrink (crenation)



What if we administer glucose

- Let's imagine we administer 5% glucose (usually named dextrose)
- Considering a molecular weight of 180g/mol
- 5%/L = 50g/L = 0,278mol/L = 278mmol/L
- Theoretically, this is isotonic… but
- $C_6H_{12}O_6$ is immediately transformed to H_2O + CO_2…. CO_2 is eliminated and therefore patient gets… WATER! (i.e. hypotonic)



Universidad de Navarra

Summary

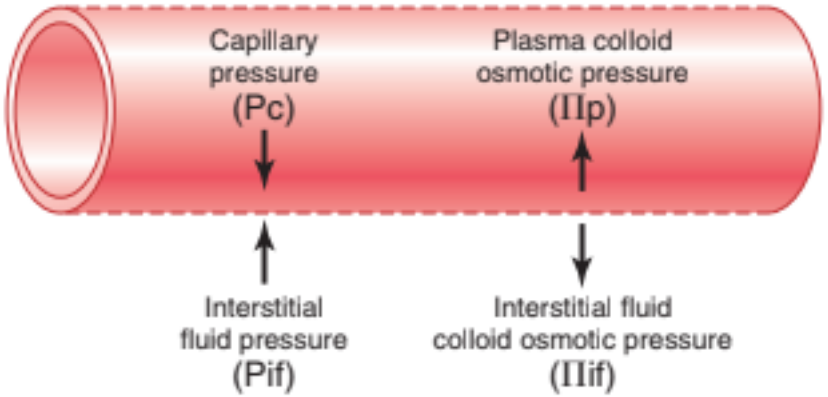
- Osmolarity, osmosis and osmotic pressure
- Aquaporins
- Plasma Osmolarity
- Osmolarity variations in ECF
- Clinical considerations: IV fluid therapy
- Water exchange within ECF (plasma and interstitial fluid)

Microcirculation

- Water and small nutrients can freely travel across endothelium
- Pores are typically up to 7nm (smaller than albumin)
- Therefore, proteins are only located in the vascular side of the endothelium, this creates an osmotic pressure called colloidosmotic (or oncotic) pressure.

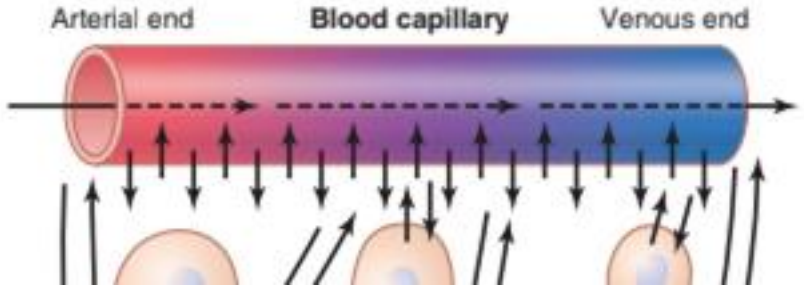
- We then get to different pressures, the hydrostatic and the oncotic

$$NFP = Pc - Pif - \Pi p - \Pi if$$



Universidad de Navarra

Microcirculation



	mm Hg
Forces Tending to Move Fluid Outward	
Capillary pressure (arterial end of capillary)	30
Negative interstitial free fluid pressure	3
Interstitial fluid colloid osmotic pressure	8
TOTAL OUTWARD FORCE	41
Forces Tending to Move Fluid Inward	
Plasma colloid osmotic pressure	28
TOTAL INWARD FORCE	28
Summation of Forces	
Outward	41
Inward	28
NET OUTWARD FORCE (AT ARTERIAL END)	13

Forces Tending to Move Fluid Inward	**mm Hg**
Plasma colloid osmotic pressure	28
TOTAL INWARD FORCE	28
Forces Tending to Move Fluid Outward	
Capillary pressure (venous end of capillary)	10
Negative interstitial free fluid pressure	3
Interstitial fluid colloid osmotic pressure	8
TOTAL OUTWARD FORCE	21
Summation of Forces	
Inward	28
Outward	21
NET INWARD FORCE	7

Mean Forces Tending to Move Fluid Outward	**mm Hg**
Mean capillary pressure	17.3
Negative interstitial free fluid pressure	3.0
Interstitial fluid colloid osmotic pressure	8.0
TOTAL OUTWARD FORCE	28.3
Mean Forces Tending to Move Fluid Inward	**mm Hg**
Plasma colloid osmotic pressure	28.0
TOTAL INWARD FORCE	28.0
Summation of Mean Forces	
Outward	28.3
Inward	28.0
NET OUTWARD FORCE	0.3



Universidad de Navarra